Microservices with Spring Boot 3 and Spring Cloud

1. Creating Microservices for Account and Loan

Aim

To create two independent Spring Boot microservices (Account and Loan) that provide REST APIs for account and loan information.

Tools Required

- Java 17
- Spring Boot 3
- Maven
- Spring Web
- Spring Boot DevTools
- Eclipse/IntelliJ IDEA
- Postman/Web Browser

Procedure

Account Microservice

1. Create a Spring Boot project using Spring Initializr.
2. Add the dependencies:
    - Spring Web
    - Spring Boot DevTools
3. Set:
    - Group: `com.cognizant`
    - Artifact: `account`
4. Create a REST Controller.
5. Implement the endpoint:

GET /accounts/{number}

6. Return a dummy account object.

Sample Response

{

"number": "00987987973432",

"type": "Savings",

"balance": 234343

}

Loan Microservice

1. Create another Spring Boot project.
2. Set:
    - Group: `com.cognizant`
    - Artifact: `loan`
3. Add Spring Web and DevTools.
4. Configure

server.port=8081

5. Implement the endpoint

GET /loans/{number}

Sample Response

{

"number": "H00987987972342",

"type": "Car",

"loan": 400000,

"emi": 3258,

"tenure": 18

}

Output

Account Service

http://localhost:8080/accounts/00987987973432

Loan Service

http://localhost:8081/loans/H00987987972342

Result

The Account and Loan Microservices were created successfully as two independent REST APIs running on different ports.

2. Create Eureka Discovery Server and Register Microservices

Aim

To create a Eureka Discovery Server and register the Account and Loan microservices with it.

Tools Required

- Spring Boot 3
- Spring Cloud Eureka Server
- Spring Cloud Eureka Client
- Maven

Procedure

Step 1: Create Eureka Discovery Server

1. Create a Spring Boot project.
2. Add dependency:
    - Eureka Server
3. Configure the application properties.

server.port=8761

eureka.client.register-with-eureka=false

eureka.client.fetch-registry=false

logging.level.com.netflix.eureka=OFF

logging.level.com.netflix.discovery=OFF

4. Add the annotation

@EnableEurekaServer

5. Run the application.
6. Open

http://localhost:8761

The Eureka dashboard should open.

Step 2: Register Account Service

1. Add dependency
- Eureka Discovery Client
2. Add annotation

@EnableDiscoveryClient

3. Configure

spring.application.name=account-service

4. Start Eureka Server.
5. Start Account Service.
6. Refresh

http://localhost:8761

The Account Service will appear in the registry.

Step 3: Register Loan Service

Repeat the same steps for the Loan Service.

Configuration

spring.application.name=loan-service

Start the Loan Service.

Refresh Eureka Dashboard.

Both services should now be registered.

Output

Eureka Dashboard

Registered Services

ACCOUNT-SERVICE

LOAN-SERVICE

Result

The Eureka Discovery Server was created successfully, and both Account and Loan Microservices were registered and discovered through Eureka.